# Lab 4 DMM Voltage and Current Measurements and Voltage-Divider Circuits

## Pre-Lab Questions

**Q1.** When measuring the resistance of a resistor or circuit, is the DMM configured as an *ohmmeter* connected in series or in parallel?

**Q2.** When measuring *resistance* using a DMM *ohmmeter*, should the component or circuit be powered or unpowered? Briefly state your reason?

**Q3.** When measuring *voltage*, is the DMM *voltmeter* connected in series or parallel with the component?

**Q4.** When measuring *current*, is the DMM *ammeter* connected in series or parallel with the circuit?

**Q5.** What must you do to the circuit before inserting the DMM to measure current?

**Q6.** Can the DMM configured as an ammeter be directly connected across a Voltage Source? What might happen? Briefly explain.

**Q7.** A 1.0 kΩ resistor and a 1.5 kΩ resistor are connected in series. What is the total resistance

**Q8.** The 1.0 kΩ and 1.5 kΩ series resistors are connected to a 10.0 V source. What is the expected circuit current?

**Q9.** What is the expected voltage drop across the 1.0 kΩ resistor?

**Q10.** What is the expected voltage drop across the 1.5 kΩ resistor?

**Q11.** What relationship should exist among the source voltage (VIN), V1 and V2?